



INTERNSHIP REPORT PRESENTATION

Tata iQ

Data Analytics Virtual Internship

The Forage — Tata Consultancy Services | Powered by Tata iQ

Prashant Sharma

Roll No. 25SCS1003004867 | B.Tech CSE

3rd Semester

Section – 2CSE37

August – September 2026



Prashant Sharma

B.Tech – Computer Science &
Engineering

ABOUT THE STUDENT

Candidate Profile



Roll Number

25SCS1003004867



Institute

IILM University



Programme

B.Tech CSE, Batch 2025–2029



Internship Domain

Tata iQ — Data Analytics



Duration

1 Months | August – September 2026

ABOUT THE PROGRAM

The Forage, Tata iQ & TCS

The Forage is a global virtual work experience platform that partners with leading companies to provide students with job-simulation programs designed to replicate real work. The Tata iQ Data Analytics Virtual Experience is delivered in partnership with Tata Consultancy Services (TCS) — one of the world's largest IT firms — and simulates the role of a data analyst embedded within Tata iQ, TCS's advanced analytics practice. The program covers the full analytics value chain for a credit risk business scenario: from EDA and predictive modelling through to stakeholder reporting and responsible AI strategy.



Industry-Aligned

Task-based curriculum mirroring
real TCS client engagements.



Tata iQ Partnership

Delivered by TCS's advanced analytics
practice.



Full Analytics Stack

EDA → Model Design → Stakeholder
Reporting → AI Strategy.

PROBLEM STATEMENT

Bridging Data & Collections Strategy at Geldium

Geldium, a fintech lending company, faced a growing delinquency crisis: customers were missing payments at an increasing rate while the collections team was operating reactively — only reaching out after a missed payment. This reactive model was proving costly, damaging to customer relationships, and unable to scale. The core analytics challenge was to identify at-risk customers before default, design a predictive model framework, and architect an AI-powered system to apply interventions proactively and at scale.

This internship addressed that challenge through a 4-task analytics workflow:



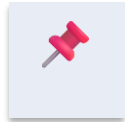
PROJECT OBJECTIVES

What the Internship Set Out to Achieve



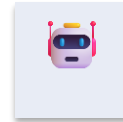
EDA & Data Quality

Profile 500-record dataset; treat missing values; standardise fields; surface risk indicators.



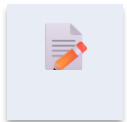
Risk Indicator Analysis

Identify correlations and top predictors of delinquency.



Predictive Model Design

Recommend XGBoost/LightGBM with LR baseline; define evaluation strategy.



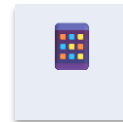
Stakeholder Reporting

Translate model insights into SMART recommendations for the Head of Collections.



Responsible AI Design

Design fair, explainable, compliant AI-powered collections system.



Executive PPT

Deliver findings as a polished executive-level PowerPoint briefing.

The Toolkit Behind the Internship



Python & Pandas

EDA, data cleaning, feature engineering



XGBoost / LightGBM

Gradient-boosted model (primary)



Logistic Regression

Transparent interpretable baseline



SHAP

Model explainability & adverse-action notices



Scikit-learn

Model training, evaluation metrics



AUC-ROC / F1 / CM

Model evaluation strategy



Microsoft Word

EDA, model plan & business reports



Microsoft PowerPoint

Executive strategy presentation

The 4-Task Analytics Journey

1

TASK 1

EDA & Data Quality

Profiled 500-customer dataset. Treated missing values (Delinquent_Account 84%, Missed_Payments 15.4%, Income 7.8%). Standardised Employment_Status. Identified top risk indicators.

2

TASK 2

Predictive Model Framework

Recommended XGBoost/LightGBM + LR baseline. Top 5 features: credit utilization, missed payments, income-to-debt ratio, account tenure, balance trend. Defined AUC-ROC, F1, Precision, Recall evaluation.

3

TASK 3

Business Summary & Stakeholder Report

Translated insights into SMART recommendation: proactive outreach for 70%+ utilization customers. Target: 15% reduction in 90-day delinquency within 2 quarters.

4

TASK 4

AI-Powered Collections Strategy PPT

Designed full AI system: 4-part workflow, agentic vs. human-in-loop table, responsible AI guardrails (SHAP, fairness, compliance), and quantitative business impact. Delivered as executive PPT.

KEY FINDINGS

EDA Insights & Top Risk Indicators



Credit Utilization

Strongest single predictor. Customers at 70%+ utilization with rising balances show markedly higher delinquency rates.

↑ Risk at >70%



Missed Payment History

Customers with 1+ late/missed payments in last 3–6 months are far more likely to become delinquent again.

Most direct behavioral signal



Income-to-Debt Ratio

Income alone does not offset risk. Capacity relative to debt is the key measure across all income bands.

Debt load > raw income



Missing Data Challenge

Delinquent_Account label was 84% missing — NOT imputed; escalated to Geldium data team. Proper governance applied.

84% target label missing

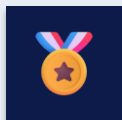
EXPECTED OUTCOMES

Skills & Capabilities Gained

- ✓ Practical ability to profile, clean, and analyze a real-world tabular dataset using EDA techniques.
- ✓ Understanding of how to design and justify a predictive model (XGBoost/LR) for financial services, including regulatory interpretability trade-offs.
- ✓ Capability to translate technical model outputs into clear, SMART business recommendations for non-technical stakeholders.
- ✓ Skill to communicate findings through a professional, executive-level PowerPoint that balances data rigour with strategic narrative.
- ✓ Knowledge of responsible AI principles in finance: fairness metrics (demographic parity, disparate impact), SHAP explainability, and compliance.
- ✓ Completed all 4 tasks and received the Tata iQ Data Analytics Virtual Internship Certificate from The Forage.

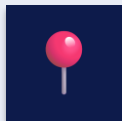
CERTIFICATION

Successfully Completed



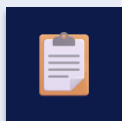
Certificate

Tata iQ Data Analytics Virtual Internship



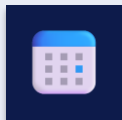
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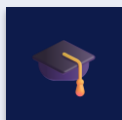
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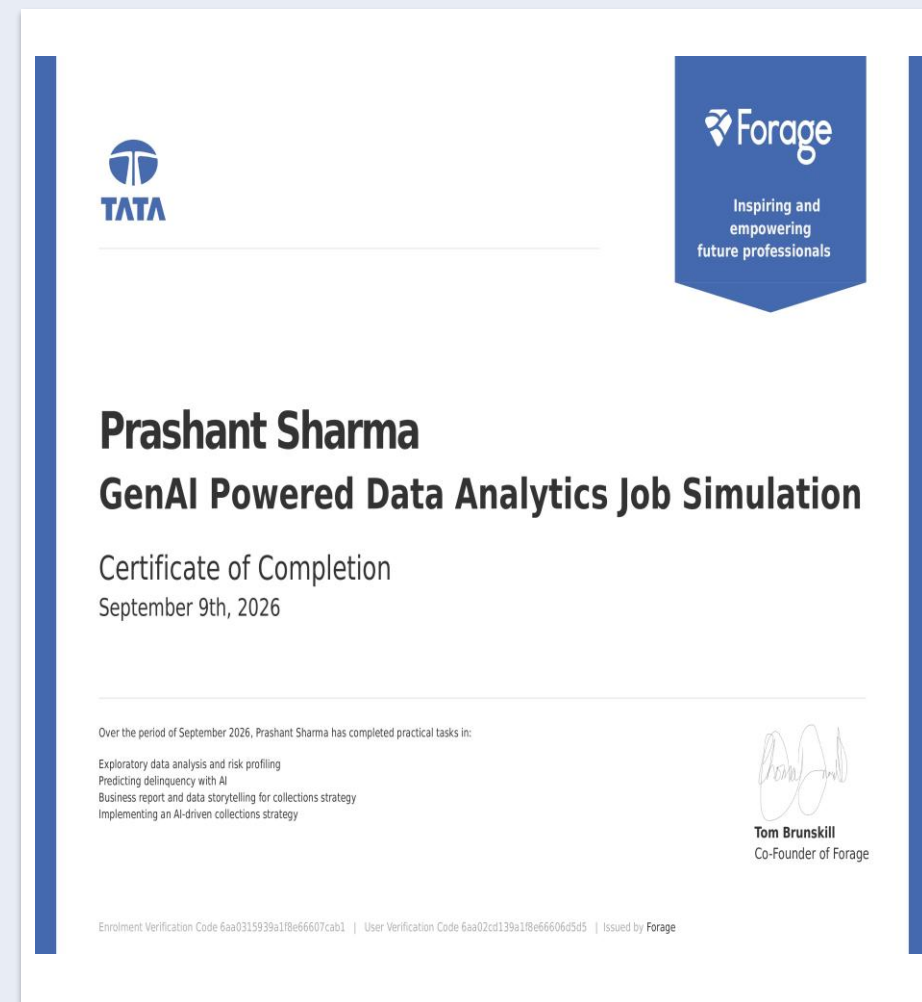
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Candidate

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Thank You

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